Khadija Korasawala

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Education

University of Illinois Urbana-Champaign, Master of Computer Science *Aug 2026 - May 2028 (Expected)*
Relevant Coursework: Artificial Intelligence, Applied Machine Learning, Advanced Data Management

Vellore Institute of Technology, B.Tech in Computer Science and Engineering *Sept 2020 – May 2024*
Relevant Coursework: Data Structures & Algorithms, Operating Systems, DBMS, Machine Learning CGPA: 9.24 / 10

Skills

Languages : Python, Java, TypeScript/JavaScript, SQL, Go, Dart, Bash
AI / ML : PyTorch, TensorFlow, Hugging Face, scikit-learn, NLP, Computer Vision, OpenCV, Generative AI, RAG, LangChain, MCP, Multi-Agent Systems
Backend & Data : FastAPI, Flask, Spring Boot, Node.js, React, Flutter, PostgreSQL, MySQL, MongoDB, Redis, Kafka, Spark, Hadoop
Cloud & DevOps : AWS (EKS, S3), GCP (GKE, Pub/Sub, Firestore), Docker, Kubernetes, GitHub Actions, CI/CD, Linux

Work Experience

Bharti Airtel Limited Pune, India
Deputy Manager – Software Developer, Full-Stack *Jun 2025 – Jul 2026*
- Fine-tuned BERT for sentiment and readability classification, achieving **88% F1** on 10K+ text samples; built a hybrid rephrasing engine combining **LLM-based generation** and rule-based validation for ambiguous content.
- Cut pricing and billing errors by **60%** by engineering validation layers across billing pipelines that verify pricing and plan data before charges are applied, safeguarding **$50M+ in annualized revenue**. (**Java, Spring Boot**)
- Owned end-to-end implementation of the redesigned **Quick Actions section** within Airtel Thanks' Postpaid experience, from frontend UI components through backend API integration, for an app serving **300M+ users**. (**React, Java, Spring Boot**)
- Resolved high-impact production incidents across payment failures, incorrect plan data, and app crashes by tracing root causes through service logs, **cutting MTTR by 15%** and curbing revenue-impacting disruptions.

Software Developer *Jul 2024 – Jun 2025*
- Built an event-labeling system tracking user actions across **50+ app features**, giving product and business teams event-level analytics on how features are used.
- Diagnosed and fixed recurring crash and data-inaccuracy defects in services supporting **300M+ customers**, shipping targeted fixes that **reduced the defect rate by 30%**.
- Implemented **API traceability** across multiple systems by adding distributed tracing headers to service requests, improving end-to-end request visibility and reliability across high-volume customer and revenue workflows.

Projects

EyeVoice – AI Accessibility App (**Flutter, Dart, Firebase, OCR, Speech APIs**) *May 2024*
- Built a Flutter accessibility app unifying **6 tools**, including OCR, AI-powered summarization, speech-to-text, and text-to-speech, into one flow that turns printed text into spoken summaries.
- Added Firebase-backed caregiver support with real-time alerts and progress tracking, earning **4.8/5 satisfaction** and a **60% reduction** in reported reading difficulty.

iMouse – Eye-Gaze Cursor Control (**Python, OpenCV, Dlib, Pynput, PyQt5**) *Jan 2024*
- Engineered a real-time eye-gaze cursor system using OpenCV and Dlib facial landmarks to track eye position and map gaze to screen coordinates, enabling fully hands-free computer control.
- Achieved **90%+ accuracy** with blink-triggered clicks detected via eye aspect ratio (EAR), with adaptive EAR and illumination thresholds and a PyQt5 settings panel for per-user calibration.

Caravanz – Personalized Travel Planner (**Random Forest, AWS Lex, JavaScript**) *Dec 2023*
- Trained a Random Forest on **2,000+ flight records** using airline, route, timing, duration, and stop information to predict cheapest fares with **90% accuracy**.
- Built a conversational trip planner on AWS Lex with a JavaScript front end that recommends personalized flights, hotels, attractions, and activities from preferences captured in chat.

Research, Achievements & Certifications

R=Research, A=Achievement, C=Certification

- **[R]:** *Terms and Conditions Checker Tool: Enhancing Digital Transparency through Automated Policy Analysis* (2023), Smart India Hackathon 2022 – built an NLP tool that analyzes terms-and-conditions documents to flag unjustified service usage, suspicious data practices, and potential privacy policy inconsistencies.
- **[A]: Airtel Ace Award** – top 2 of 50 for the fine-tuned BERT sentiment and readability classification project and overall performance; **CodeKaze** – College Rank 3; **GirlScript Winter Open Source** – 5+ PRs merged.
- **[C]:** Algorithms Specialization (Stanford), Applied ML in Python (UMich), OCI AI Foundations Associate, AWS Cloud Practitioner.